Project Design Phase-II
Technology Stack (Architecture & Stack)

Date	29/09/2026
Team ID	SWTID-2026-2908
Project Name	FitSense – AI-Powered Fitness Recommendation System
Maximum Marks	4 Marks

Technical Architecture:

The FitSense backend follows a modular architecture with REST APIs. User requests pass through the Express.js server, authentication and routing, controllers/services, and then interact with MongoDB or Google Gemini AI before returning JSON responses.

Component	Description	Technology
User / API Client	Sends registration, login, workout and AI recommendation requests	REST + JSON
Application Server	Handles routing, controllers and services	Node.js + Express.js + MVC
Authentication	Validates user access and protects routes	JWT
Password Security	Hashes user passwords	bcrypt.js
Database	Stores users and workout records	MongoDB + Mongoose
AI Interface	Generates personalized plans and fitness insights	Google Gemini API
API Testing	Tests REST endpoints and responses	Postman

Table-1: Components & Technologies

S.No	Component	Description	Technology
1	User Interface / API Client	How user interacts with application	REST API / JSON; Postman for testing
2	Application Logic-1	Authentication and API routing	Node.js + Express.js
3	Application Logic-2	Application business logic and CRUD operations	Express.js + MVC Controllers/Services
4	Application Logic-3	AI recommendation and fitness insights	Google Gemini
5	Database	User and workout data storage	MongoDB + Mongoose
6	Cloud Database	Database service on cloud (if deployed)	MongoDB Atlas
7	File Storage	No dedicated file-storage requirement in the current scope	Not specified
8	External API-1	AI-powered fitness recommendations	Google Gemini API
9	External API-2	No additional external API in current scope	Not specified
10	Machine Learning Model	Personalized workout plans and fitness insights	Google Gemini AI
11	Infrastructure (Server / Cloud)	Application deployment	Local Node.js server / cloud deployment as applicable

Table-2: Application Characteristics

S.No	Characteristics	Description	Technology
1	Open-Source Frameworks	Frameworks used for backend development	Node.js, Express.js, Mongoose
2	Security	Authentication and password protection	JWT, bcrypt.js
3	Scalable Architecture	Modular separation of models, controllers and services supports expansion	MVC, REST API, MongoDB
4	Availability	Backend and database can be deployed as services for continuous access	Node.js/Express.js + MongoDB Atlas
5	Performance	Efficient REST APIs, structured database access and modular processing	Express.js + MongoDB + JSON